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Original article

Insights into the aeroelastic stability of a wing with distributed electric propulsion systems

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ABSTRACT

This paper presents a comprehensive aeroelastic stability analysis of a wing featuring a Distributed Electric Propulsion (DEP) system, addressing key challenges in the design and evaluation of next-generation aircraft, such as electric and hybrid-electric vehicles. The study examines a benchmark cantilever wing equipped with thrusting propellers, varying their number and spanwise positioning from one to five while maintaining consistent propeller disk loading and total mass. Two numerical frameworks are employed for aeroelastic assessments. A frequency-domain method is used to generate flutter curves and identify flutter onset, while mid-fidelity simulations based on coupled multibody dynamics and vortex particle methods refine the flutter point while capturing the complex aerodynamic interactions between the propellers and the wing. The analysis focuses on airplane-mode flight conditions, excluding propeller-pylon failures. Across all configurations examined, wing flutter emerged as the primary instability mechanism, regardless of propeller placement. The most stable configuration featured a single propeller positioned at the wingtip, whereas increasing the number of propellers led to a reduction in flutter speed. Aerodynamic interactions further decrease flutter speed, with thrust conditions promoting destabilization compared to windmilling scenarios. Notably, configurations with multiple propellers exhibit greater susceptibility to these effects. Finally, design guidelines are proposed to enhance the stability of standalone propeller-pylon assemblies and wings equipped with DEP systems.

1. Introduction

The aviation industry is under increasing pressure to adopt efficient and cost-effective solutions due to growing air travel demand and stringent environmental regulations. Inspired by the automotive sector's shift to electrification, aviation is exploring electrified propulsion systems to reduce CO2, NOx, and noise emissions. Key climate targets, such as the "Flightpath 2050" [1] and "Destination 2050" initiatives [2], aim for net-zero CO2 emissions by 2050 through the introduction of sustainable aviation fuels and electric aircraft technologies.

A promising innovation is the Distributed Electric Propulsion (DEP), which uses lightweight electric motors distributed across the airframe to enhance efficiency and design flexibility [3–7]. However, the low energy density of batteries and the need for mass-optimized structures present significant challenges. Reduced structural stiffness can lead to increased dynamic loading and aeroelastic instabilities, such as flutter and whirl flutter, which pose critical safety risks. Certification regulations require a thorough demonstration of stability across the aircraft's operational envelope to ensure safe operation (e.g., see paragraph CS 25.629 in [8] or CS 23.2245 in [9]).

Predicting and evaluating aeroelastic phenomena in conventional aircraft typically involves computational methods or experimental studies. Flutter tests are costly and complex, while computational methods vary in accuracy depending on the design phase [10]. Reduced-order models are commonly used in the preliminary design phase due to their balance between efficiency and accuracy, enabling the early identification of unstable designs. However, high-fidelity time-domain simulations, which offer more precise results, are reserved for detailed design and certification stages due to their high computational cost.

The rise of DEP systems introduces new aeroelastic challenges that are not yet fully understood, particularly regarding the distribution of propellers along the wing span. The lack of experience with these configurations makes it difficult to develop suitable reduced-order models for early-stage design assessment. Therefore, further research is needed to explore the aeroelastic behavior of DEP systems and establish efficient methods for evaluating stability during the aircraft design process.

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Despite the recent developments of eVTOL aircraft, the literature on the aeroelastic stability of DEP systems is limited. Hoover and Shen [11] analyzed the whirl flutter stability of the NASA all-electric experimental aircraft, the X-57 Maxwell, under free-flight conditions. The X-57 features 14 propellers, including two large outboard propellers active during all flight phases and twelve smaller inboard propellers that assist during take-off and landing. The study demonstrated that the fundamental wing modes remain stable up to twice the expected cruise speed. However, propeller whirl flutter was observed only when the pylon spring stiffness was significantly reduced to simulate a severely damaged pylon mount. Amoozgar et al. [12] analyzed a configuration similar to the X-57 Maxwell to investigate the impact of DEP systems on wing stability. Their findings revealed that the wing equipped with DEP exhibited greater stability compared to the clean wing configuration. The study highlighted that the wingtip propulsor had the most significant influence on both the flutter speed and frequency, effectively increasing stability margins, whereas the high-lift motors had a negligible effect on the wing's stability. Gul and Yeo [13] developed a family of generic, hingeless hub-distributed propulsion tiltrotor configurations to investigate whirl flutter instability across a range of design variables. Their study employed a systematic approach to assess the stability of distributed propulsion configurations and provided guidelines to designers, though the models were based on rotors rather than propellers. Recently, Böhnisch et al. investigated aeroelastic phenomena in a slender wing with a single propeller, focusing on the transition from wing flutter to whirl flutter and the underlying instability mechanisms [14]. The present study extends that analysis to configurations with multiple propellers, specifically considering DEP layouts.

The primary objective of this work is to assess how critical speeds and instability mechanisms vary across different DEP configurations. The number of propeller-pylon systems ranges from one to five, and the influence of propeller positioning on flutter speed is examined to identify potentially advantageous layouts. The analyses are performed using a linear frequency-domain tool, offering key insights into flutter points, natural frequencies, damping ratios, and mode shapes through a robust and efficient formulation. Subsequently, aerodynamic interference effects between the wing and the propellers are taken into account using a vortex-particle method (VPM). Aeroelastic analyses are performed by coupling the VPM with a multibody dynamics tool, which considers the nonlinear dynamics of the interconnected rigid and flexible bodies representing the aircraft components. Based on the findings, the study provides recommendations for flutter and whirl flutter clearance of aircraft with DEP systems and offers design guidelines to enhance aeroelastic stability margins.

The paper begins by introducing the numerical frameworks used to assess aeroelastic stability in wing–propeller systems (Sec. 2). Next, multiple DEP-inspired configurations are presented, along with a detailed description of the propeller scaling methodology (Sec. 3). The impact of propellers' number and positioning on flutter onset is then evaluated, remarking on the effects of aerodynamic interference (Sec. 4). Design guidelines for DEP aircraft are subsequently derived (Sec. 5). Finally, the conclusions (Sec. 6) summarize the main findings and propose directions for future research.

2. Numerical frameworks

2.1. SDBox

A robust, fast, and efficient methodology is essential for conducting rapid aeroelastic stability analyses in the context of large parameter studies. The frequency-domain approach is well-suited for this purpose, as it enables the reduction of aeroelastic equations through an appropriate modal basis while facilitating efficient stability assessments. To systematically investigate aeroelastic stability in aircraft wings with multiple propellers, the authors developed a MATLAB toolbox called SDBox [14].

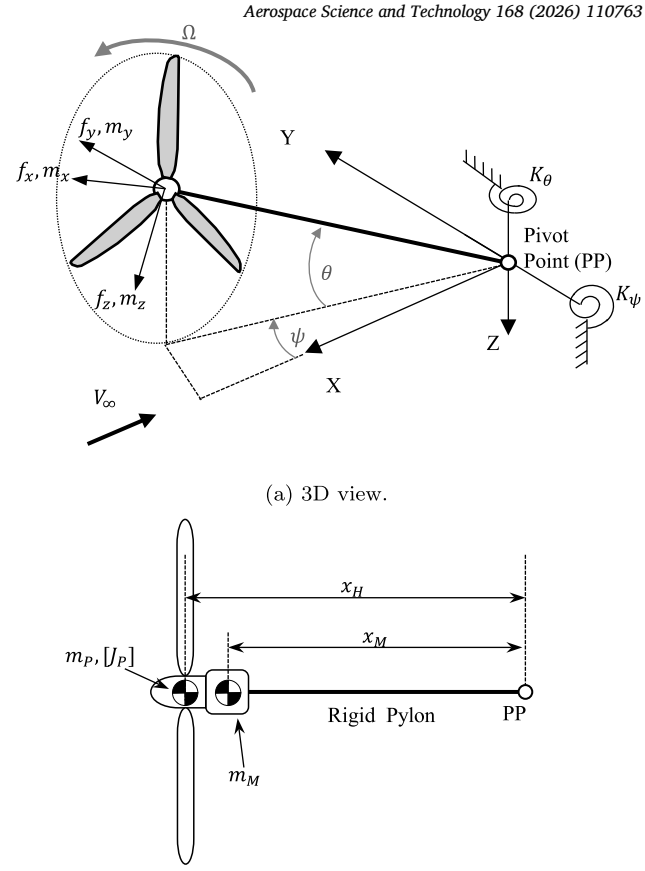


Fig. 1. Propeller-pylon model.

The wing, as the primary structural component, is modeled using beam elements based on Timoshenko–Ehrenfest theory. This formulation accounts for shear deformation by assuming a constant shear stress distribution across the beam's cross-section. However, due to this assumption, warping effects are generally neglected. Unsteady aerodynamic forces on the wing are computed using the Doublet Lattice Method (DLM) as a function of the reduced frequency and Mach number, following the formulation proposed by Albano and Rodden [15].

Propellers are mounted to the wing via rotational springs, representing a simplified, flexible pylon system that permits pitch (θ) and yaw (ψ) rotations about the pivot point. The propeller rotational speed is assigned (Ω). A sketch of the propeller-pylon system is shown in Fig. 1.

The basic propeller-pylon model was developed in the 1960s for whirl flutter predictions, through the foundational work of researchers such as Houbolt, Reed, and Bland [16–18]. Each component within the assembly, including the blades, is assumed to be rigid. This assumption provides a conservative estimate for whirl flutter boundaries [19]. Mass and inertia properties are modeled using lumped masses. The propeller disk is located at a distance x_H from the pivoting point. The propeller mass m_P and the corresponding inertia tensor $[J_P]$ are treated as rotating masses. The lumped mass of the electric motor, m_M, positioned at a distance x_M from the pivoting point, is also considered and is referred to as the non-rotating mass group. The rotational springs are denoted as K_θ and K_ψ. Propeller aerodynamics are modeled through stability derivatives as described in [16,20]; however, instead of using an analytical formulation for the derivatives, an enhanced, fully unsteady approach is applied [21,22]. The open-source tool DUST [23] is exploited for the generation of propeller unsteady aerodynamic forces. Propellers are modeled in DUST with lifting line elements, defined using the twist and chord parameters at each section. Airfoil aerodynamic coefficients are input using the .c81 file format, over the expected Mach

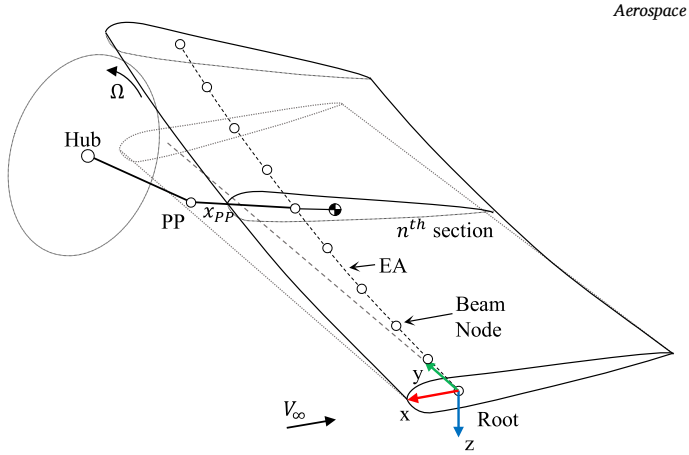


Fig. 2. General coupled wing-propeller system.

and Reynolds number ranges for the operating envelope. To obtain the unsteady aerodynamic forces, the propeller is first trimmed for a specific flight condition and then harmonically excited in pitch at its pivot point with different excitation frequencies. The time histories of the forces and moments at the propeller hub (see Fig. 1a) are then measured to determine the stability derivatives by performing an FFT and computing the transfer function at the given excitation frequency. Due to the propeller symmetry, the aerodynamic stability derivatives are computed only by exciting the pitch motion. The steady aerodynamic torque is incorporated in the propeller dynamics, according to the recent work proposed by Böhnisch et al. [24], providing a stabilizing effect on whirl flutter when considering powered propellers.

The two subsystems, i.e. the wing and the propeller, are coupled using Kane's method [25], with the user defining the distance between the pivot point and the wing's elastic axis, x_{PP}, as illustrated in Fig. 2. The mathematical formulation can be readily extended to accommodate multiple propellers. The coupled aeroelastic system is described by the classical flutter equations and solved using the $p-k$ method (see Chapter 9, Ref. [26]). SDBox has demonstrated high efficiency and accuracy in rapidly predicting flutter speeds and was successfully validated in [14].

2.2. MBDyn-DUST

A coupled toolchain, comprising MBDyn [27] and DUST [23], is employed to account for aerodynamic interference between the propellers and the wing in aeroelastic stability predictions. MBDyn is an open-source, general-purpose multibody dynamics solver capable of modeling complex mechanical systems. It simulates the nonlinear dynamics of rigid and flexible bodies interconnected by kinematic constraints. The equations of motion are formulated in differential-algebraic form and numerically solved using various A/L-stable integration methods, offering robust and accurate solutions for dynamic simulations. DUST, which was introduced in the previous section for the generation of the propeller aerodynamic forces, is an aerodynamic solver based on potential flow theory. It employs a free-wake representation with vortex particles, enabling accurate modeling of complex geometries. The tool offers multiple elements, including lifting lines for slender lift-generating surfaces, vortex lattice elements, and surface panels. The vortex particle method provides a stable description of the free-vorticity flow field, particularly suited for configurations characterized by strong aerodynamic interactions. This Lagrangian grid-free approach eliminates the need for a volume mesh of the surrounding flow, efficiently capturing wake interactions produced by lifting surfaces and bodies, as commonly observed in rotary-wing vehicle applications. To mitigate the computational costs associated with vortex particle interactions, a Cartesian Fast Multipole Method (FMM) is employed [28,29]. The coupling between MBDyn and DUST is facilitated through the open-source coupling library PreCICE [30,31]. PreCICE implements a tight coupling scheme, enabling the seamless exchange of information between the two solvers. This framework enhances both the accuracy and efficiency of the simulations by enabling consistent interaction between the multibody dynamics and aerodynamic solvers. The coupled MBDyn–DUST framework has been widely utilized for aeroelastic analyses of both fixed-wing and rotary-wing aircraft, including tiltrotors [32,33]. This approach was chosen by the authors for its computational efficiency over high-fidelity CSD-CFD tools, while also allowing for the inclusion of aerodynamic interference in aeroelastic studies.

An overview of the wing-DEP model setup in MBDyn and DUST is given in Fig. 3.

The wing's elastic properties are modeled in MBDyn using three-node beam elements [34]. The wing is clamped at its root. The mass and inertia properties are modeled using lumped masses at each node, accounting for the offset between the elastic axis and the center of gravity. Propellers can be attached at any spanwise location on the wing. The pivoting point of the propeller is rigidly connected to a specific beam node, towards the leading edge. The propeller hub is connected to the pivoting point through two joints: a total joint and a deformable hinge. The deformable hinge introduces the pylon rotational stiffnesses in yaw and pitch. Damping effects are neglected on this joint. The propeller node is located at the same position as the hub node. They are both connected via an axial rotation constraint, which enforces the rotation of the propeller. Additionally, multiple nodes are rigidly attached to the propeller node, representing the blades. The mass of the motor is connected to the hub node (non-rotating mass). Propeller blades are assumed to be rigid. The corresponding mass/inertia is lumped at the blade center of mass. The dynamic nodes of the propeller blades are selected for coupling with DUST. Propeller aerodynamics in DUST are modeled using lifting line elements on each blade, as discussed in the previous section. Surface panels are employed to represent the wing's geometry. To accurately capture the curvature of the leading edge, the chord distribution is discretized using a half-cosine function (see Fig. 4). The number of panels defined on the wing and lifting line elements on the propeller blades has been selected to capture the wing and propeller performance in steady conditions. A mesh-independent study was conducted to ensure the convergence of lift/moment coefficients for the wing and thrust/torque for the propeller.

Aeroelastic stability analyses of individual components conducted using MBDyn-DUST demonstrated excellent agreement with SDBox. Both tools successfully predicted the same flutter mechanisms for the isolated wing and the same whirl flutter behavior for the standalone propeller [14]. The validation of the coupled wing-propeller system was also examined in [35], leveraging the hybrid approach of MBDyn-DUST, which enables independent modeling of wing and propeller aerodynamics while neglecting interference effects. The strong correlation between the two numerical frameworks supports the use of MBDyn-DUST to quantify the impact of aerodynamic interference on instability mecha-

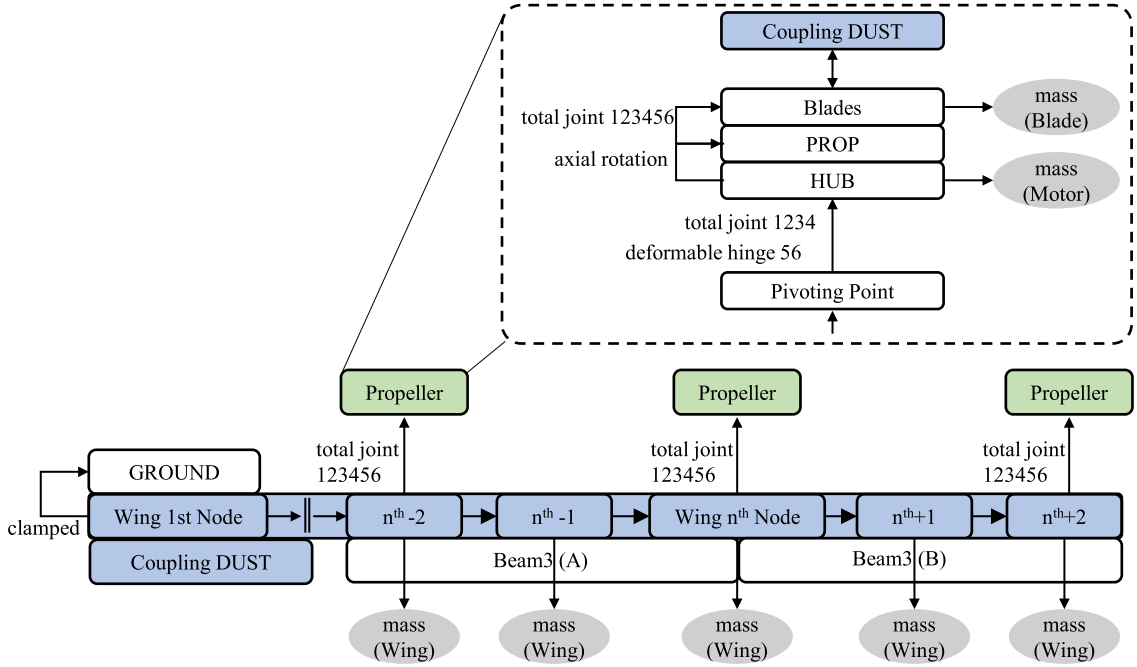


Fig. 3. Sketch of the coupled wing-DEP model in MBDyn–DUST.

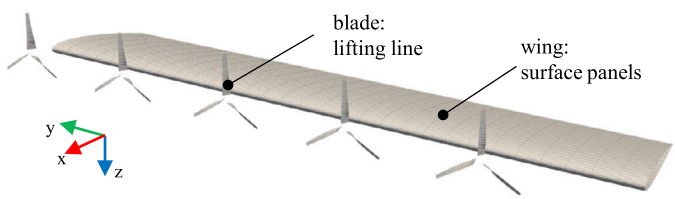


Fig. 4. DUST aerodynamic model for wing and propellers.

Table 1
Aircraft general specifications.

Description	Value	Units
Maximum Take-off Mass, $MTOM$	2800	kg
Cruise speed, V_{cr}	72	$\mathrm{m\,s^{-1}}$
Lift-to-Drag ratio, $\frac{L}{D}$	11	–
Thrust required (at cruise), T_{req}	2500	N

Table 2
Wing structural properties.

Description	Value	Units
Semi-span length	5.7	m
Chord, c	1.25	m
Elastic axis, $\%c$	35	–
Center-of-gravity axis, $\%c$	51	–
Mass/unit span	25.0	$\mathrm{kg\,m^{-1}}$
Inertia/unit span (about CG)	4.5	kg m
In-plane stiffness, EI_{zz}	1.16×10^7	$\mathrm{N\,m^2}$
Out-of-plane stiffness, EI_{xx}	2.90×10^6	$\mathrm{N\,m^2}$
Torsional stiffness, GJ	2.70×10^5	$\mathrm{N\,m^2}$

nisms in wing-DEP systems, provided that both the structural and aerodynamic behaviors remain linear.

3. Configuration layouts

A cantilever wing with propellers distributed along the span is considered in the following. The propellers are positioned ahead of the leading edge in a tractor configuration, with their axes aligned with the freestream velocity, ensuring axial flow conditions. Due to symmetry, only the semi-span wing is modeled, while other aircraft components (e.g., fuselage and empennages) are excluded from the analysis.

The analyzed configurations are based on a mid-range, small aircraft with a Maximum Take-Off Mass (MTOM) of 2800 kg. The basic aircraft specifications are given in Table 1. Standard sea-level conditions are assumed for all simulations.

The wing is rectangular and the corresponding properties are given in Table 2. The wing model also includes structural damping through a stiffness-proportional damping factor equal to 0.1%. The eigenfrequencies and damping ratios of the isolated wing in a vacuum computed by SDBox are listed in Table 3. The corresponding mode shapes are shown in Fig. 5. Flutter and divergence velocities of the isolated wing obtained in SDBox are 82.6 $\mathrm{m\,s^{-1}}$ and 199.3 $\mathrm{m\,s^{-1}}$, respectively. The corresponding flutter frequency is 7.8 Hz. The flutter mechanism is triggered by the first two modes, leading to a classical bending-torsion coupling.

Table 3
Eigenfrequencies and damping ratios of the cantilever wing in a vacuum (SDBox).

Mode	Description	Freq., Hz	Damp., %
1	1st Out-of-plane bending	5.59	1.8
2	1st Torsion	9.48	3.0
3	1st In-plane bending	11.69	3.7
4	2nd Torsion	24.55	7.7

Five distinct configuration layouts are proposed, varying primarily in the number of propellers per semi-span wing, ranging from one to five. All configurations utilize identical propellers. The geometric and structural design parameters governing the attachment of the propellers

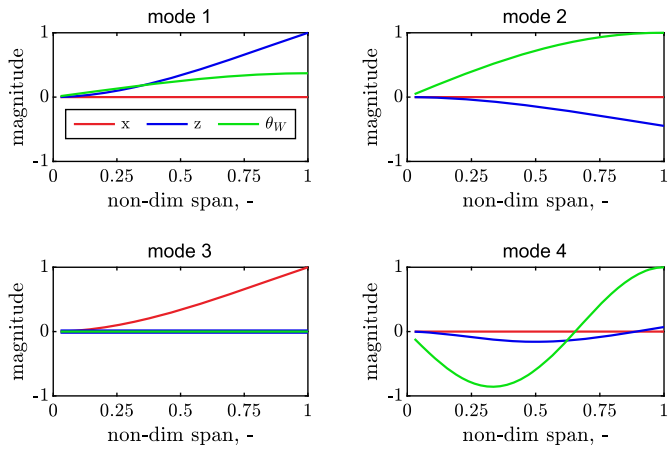


Fig. 5. Mode shapes of the cantilever wing in a vacuum (SDBox). (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

Table 4
Propeller design variables.

Description	Symbol	Value
Number of propellers	N_P	1-5
Semi-span positions	$\eta_{P,i}$	variable
Pivoting Point position	$x_{PP,i}$	0.3 m
Motor position	$x_{M,i}$	0.3 m
Hub position	$x_{H,i}$	0.4 m
Spring stiffness	$K_{\theta,i} = K_{\psi,i}$	constant

Table 5
Propeller diameters.

N_P	Thrust per propeller T_P, N	Diameter D, m
1	1250	1.78
2	625	1.26
3	417	1.03
4	313	0.89
5	250	0.80

to the wing are summarized in Table 4. Of course, the total number of design variables depends on the number of propellers. To minimize complexity, a few parameters are constrained. The chordwise position of the pivot point is fixed at 0.3 m ahead of the elastic axis and remains consistent across all configurations. Additionally, the pivot point's spring stiffnesses are identical in both pitch and yaw direction and fixed to a constant value (i.e., isotropic conditions).

The number and distribution of propellers along the semi-span vary. To prevent interference between adjacent propellers, a minimum spacing of $\delta_P \cdot D$ (measured from hub to hub) is imposed, where D is the propeller diameter and δ_P is a scaling factor greater than one. In this study, δ_P is set to 1.4.

Five propellers must be designed, one for each configuration investigated. To ensure consistency across configurations while preventing excessively high blade aspect ratios, a constant disk loading of 500 N m^{-2} is applied to all propellers. Based on the aircraft's thrust requirement during the cruise, the diameters of the five propellers can be easily determined (see Table 5).

The various configuration layouts are illustrated in Fig. 6. The propellers are arranged in close alignment, maintaining the minimum required spacing. As previously mentioned, the spanwise position remains a design variable, and its influence on aeroelastic stability will be discussed in Section 4.

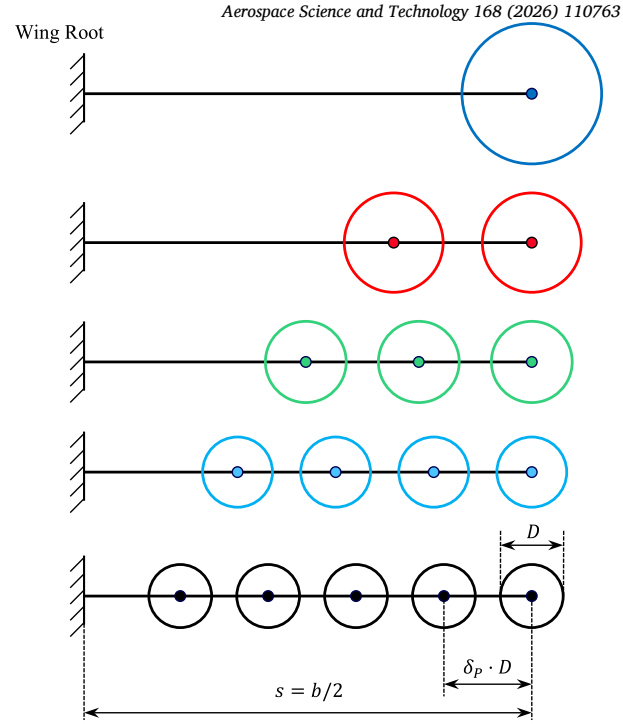


Fig. 6. Front view of different configurations under investigation.

Table 6
Propeller input parameters for MIL design.

Parameter	Units	Number of Propellers				
		1	2	3	4	5
D	m	1.78	1.26	1.03	0.89	0.80
N_b	–	3	3	3	3	3
Ω	rpm	2524	3569	4371	5047	5643
V_{cr}	m s^{-1}	72	72	72	72	72
T_P	N	1250	625	417	313	250

3.1. Propellers' scaling for DEP systems

The propellers are designed using an adapted Minimum-Induced-Loss (MIL) methodology, originally developed by Adkins and Liebeck [36] and later refined by Bergmann [37] within the MATLAB toolbox PropCODE. This toolbox, developed at FH Aachen, facilitates the structural, aerodynamic, and aeroacoustic design of modern propellers [38, 39].

The key input parameters for the MIL design include the propeller diameter (D), number of blades (N_b), and the design point, which is defined by rotational speed (Ω), cruise speed (V_{cr}), and thrust requirement (T_P). Most of these parameters have already been established and are summarized in Table 6 for reference. For consistency, all propellers are designed with three blades. The flight speed and thrust requirements are defined by the aircraft's general specifications. The rotational speed is determined to maintain a blade tip Mach number of 0.725, as calculated using Eq. (1).

$$\Omega = \sqrt{\frac{(M_{Tip} \cdot a)^2 - V_{cr}^2}{\left(\frac{D}{2}\right)^2}}, \tag{1}$$

where a represents the speed of sound, computed under sea-level conditions.

The propeller blade design is based on 12 radial sections, with the Clark Y airfoil used consistently across all sections. As a result, PropCode generates the final blade geometry, including chord and twist distributions. Additionally, performance curves are provided for vari-

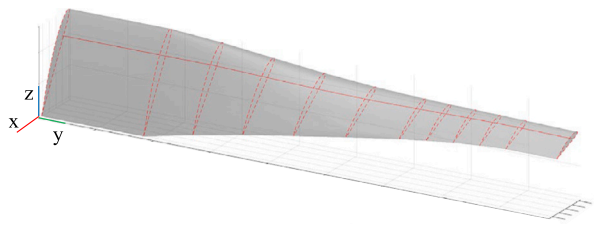
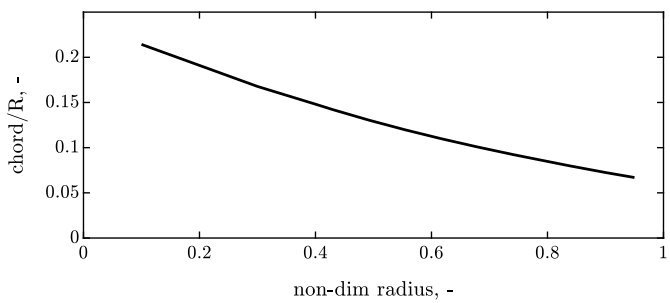
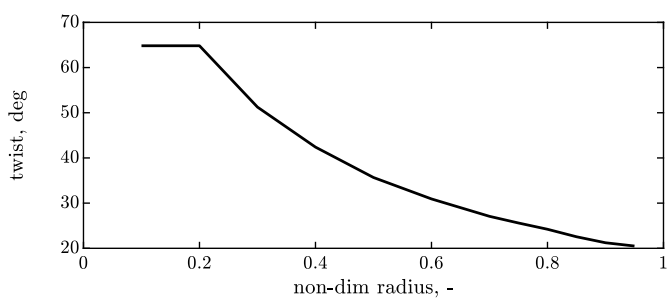


Fig. 7. 3D blade design.



(a) Normalized chord distribution.



(b) Twist distribution.

Fig. 8. Representative blade geometry distributions.

ous operating conditions. A 3D visualization of the blade is shown in Fig. 7.

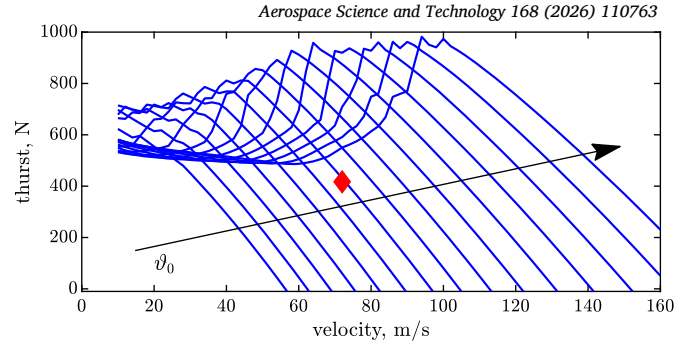
The normalized chord and twist distributions are presented in Fig. 8. Due to the imposed constraints, which ensure constant disk loading and radial velocities, these distributions remain identical for all propellers.

Performance curves for varying collective pitch angles are given in Fig. 9, exemplary for the propeller with a diameter of 1.03 m. The red diamond marks the design condition, representing the thrust required during cruise.

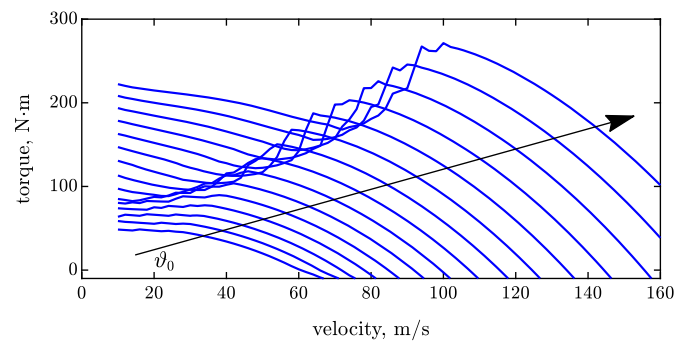
The windmilling condition (zero torque) is initially used to generate the propeller aerodynamic database in DUST, as it represents the most critical scenario for propeller whirl flutter [17]. The corresponding trim chart, which correlates collective pitch with freestream velocity under windmilling conditions, is shown in Fig. 10. Due to the selected constraints, the trim chart remains nearly identical for all propellers, with only minor variations resulting from numerical rounding. Notably, the freestream velocity can be directly related to the advance ratio, as changes in rotational speed and propeller diameter are accounted for (see Table 6).

The propeller aerodynamic database is constructed using 12 freestream velocities, indicated by circular markers in Fig. 10. For each velocity, aerodynamic derivatives are computed at five different reduced frequencies, namely:

$$k = \frac{\omega \cdot \ell}{V_\infty} = 0.03, 0.05, 0.1, 0.3, 0.5, \tag{2}$$



(a) Thrust.



(b) Torque.

Fig. 9. Performance curves of propeller #3 at different collective pitch angles ($-12° \leq \vartheta_0 \leq 16°$, $\Delta\vartheta_0 = 2°$) with constant rotational speed, $\Omega = 4371$ rpm.

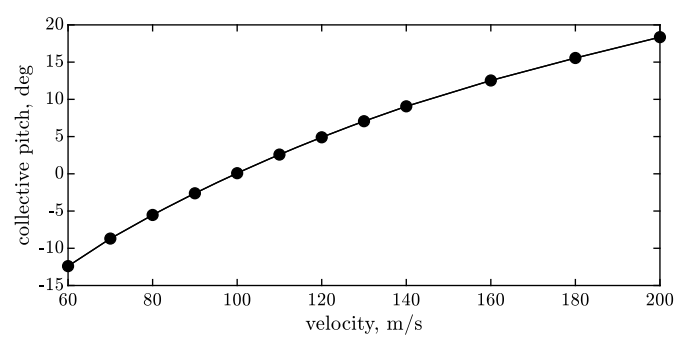


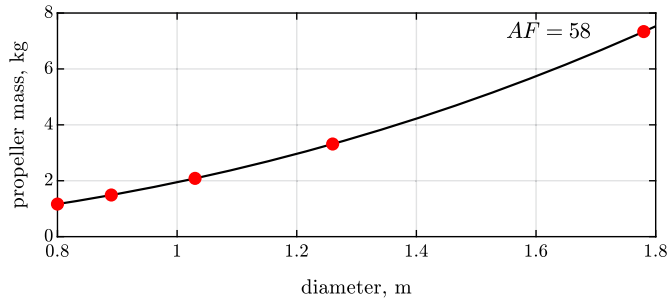
Fig. 10. Collective pitch required for windmilling condition as a function of the freestream velocity.

where the reference length ℓ is half of the propeller radius. In total, 60 time-domain simulations were conducted in DUST for each propeller, forming the basis of the aerodynamic derivative database required by SDBox.

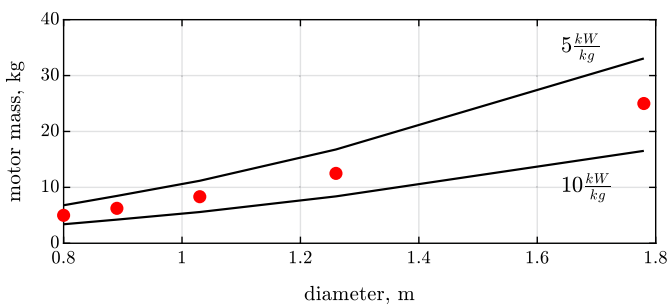
Finally, the mass and inertia properties of the propellers and motors are evaluated. The total mass (m_T) is constrained to remain constant to maintain consistency across all investigated configurations. However, while the total mass remains unchanged, the mass distribution — and consequently, the inertia — varies across the five configurations, potentially influencing the system's dynamic behavior.

$$m_T = N_P \cdot (m_P + m_M) \overset{!}{=} const. \tag{3}$$

The propeller mass is estimated using an empirical formula proposed by Amatt et al. [40] for hollow fiberglass blades, which expresses mass as a function of the propeller diameter and the activity factor (AF). The activity factor, a dimensionless parameter proportional to the ratio of the propeller's mean chord to its diameter, is equal to 58 for all five propellers examined in this study. The selected propeller masses follow



(a) Propeller.



(b) Motor.

Fig. 11. Selected masses (red dots) for propellers and motors.

the curve derived from the empirical model as a function of propeller diameter and are shown in Fig. 11a. The propeller inertia tensor, $[J_P]$, is subsequently computed by considering only the contribution of the blades, which are modeled as uniform rods.

The motor masses are selected to satisfy the total mass constraint. According to [41], electric motors exhibit a power-to-mass ratio between $5\,\mathrm{kW\,kg^{-1}}$ and $10\,\mathrm{kW\,kg^{-1}}$. In this study, the maximum power requirement per motor is estimated based on the cruise power (P_{cr}) obtained from the performance curves, assuming that only 65% of the total power is utilized during this flight phase. Using this assumption, the motor mass boundaries can be determined using Eq. (4), as follows:

$$m_{M,min} = \frac{P_{cr}}{0.65 \cdot 10}, \tag{4a}$$

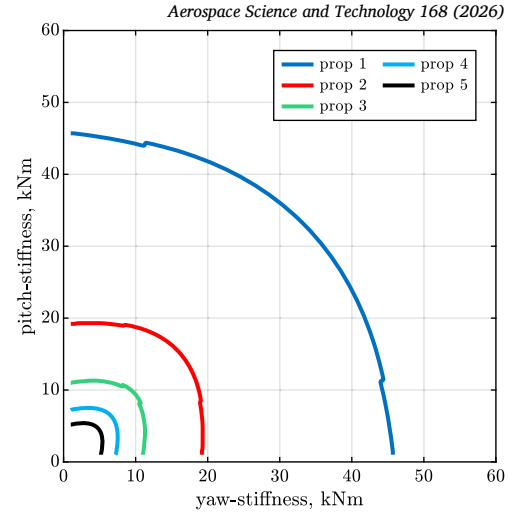
$$m_{M,max} = \frac{P_{cr}}{0.65 \cdot 5}. \tag{4b}$$

Notably, the chosen motor masses (i.e., red dots in Fig. 11b) fall within the boundaries defined by Eq. (4), making them a representative estimate for next-generation electric propulsion systems.

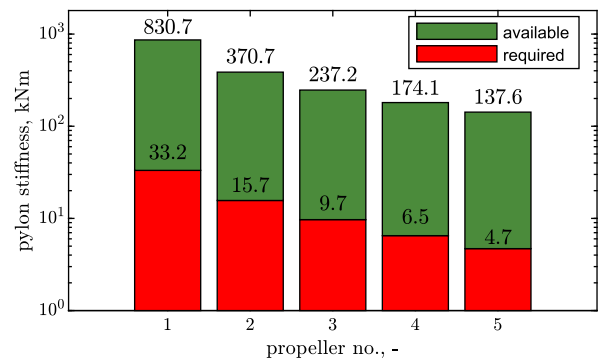
3.2. Whirl flutter boundaries

In the following, whirl flutter analyses in windmilling conditions are presented for each isolated propeller, with the freestream velocity set to cruise speed. The corresponding stability maps, as a function of the rotational spring stiffness in pitch and yaw, are given in Fig. 12a. Whirl flutter occurs for stiffness combinations within the stability boundaries, while points beyond these curves indicate a stable response. The minimum required stiffnesses in isotropic conditions are compared across configurations in Fig. 12b (orange bars). As propeller size decreases, the required stiffness for stability also reduces. This aligns with expectations, as larger propellers are more susceptible to whirl flutter when operating under the same flight conditions.

Under nominal design conditions, it is generally recommended to maintain adequate frequency separation between the natural frequencies of the isolated propeller-pylon system and the wing's elastic modes critical for flutter to mitigate resonance phenomena. Consequently, the pivot point's spring stiffnesses are designed so that the natural frequen-



(a) Whirl flutter stability maps.



(b) Minimum required vs available stiffness.

Fig. 12. Whirl flutter boundaries for isolated propellers in windmilling conditions at V_{cr}.

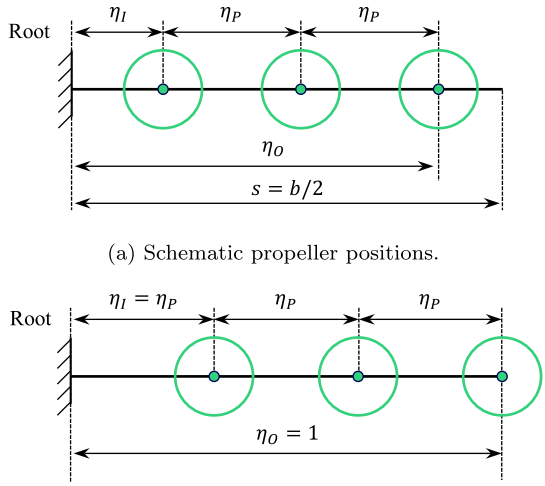
cies of the isolated propeller-pylon system exceed the highest elastic frequencies of the isolated wing, which could potentially induce flutter. In this work, a non-rotating propeller frequency of $f_P = 70\,\mathrm{Hz}$ is selected. This frequency is seven times higher than the natural frequency of the isolated wing's flutter mode 2 (9.48 Hz) and nearly three times higher than that of mode 4 (24.55 Hz). The choice of the non-rotating propeller frequency allows us to determine K_θ and K_ψ for the different configuration layouts, which satisfy the minimum stiffness requirement for the whirl flutter of the standalone propeller, as shown in Fig. 12b (green bars). This condition also ensures that the whirl flutter speeds for the five propeller-pylon systems in windmilling conditions are higher than the flutter speed of the isolated wing.

4. Influence of DEP system on wing's aeroelastic stability

4.1. Position and spacing of DEP system

Based on the configuration layouts, multiple spanwise positions for the propellers are possible. Configurations with fewer propellers allow for greater flexibility in placement compared to those with a higher number of propellers. In the following, the three-propeller configuration is used as a reference. Fig. 13a illustrates the key parameters defining the spanwise positioning of the DEP system.

Each distance is expressed as a non-dimensional value relative to the wing semi-span. To reduce the number of possible configurations, it is assumed that the spacing between propellers (η_P) remains equal but variable. The position of the outer propeller (η_O) is specified, allowing the position of the inner propeller (η_I) to be determined accordingly. It



(a) Schematic propeller positions.

(b) Equally spaced configuration.

Fig. 13. Distances used to describe the propeller positions.

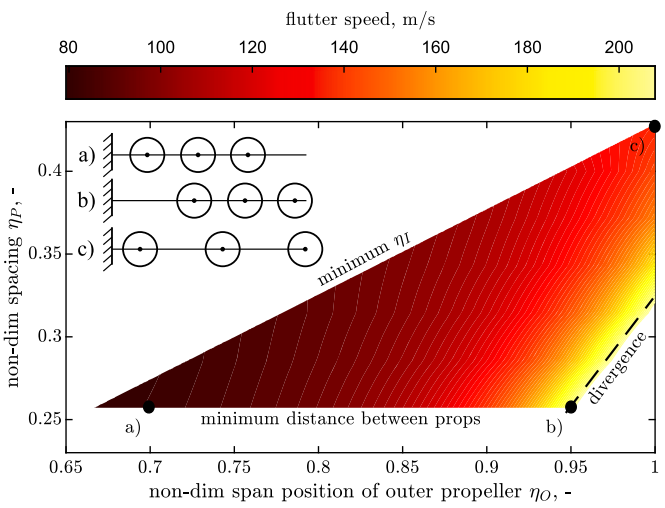


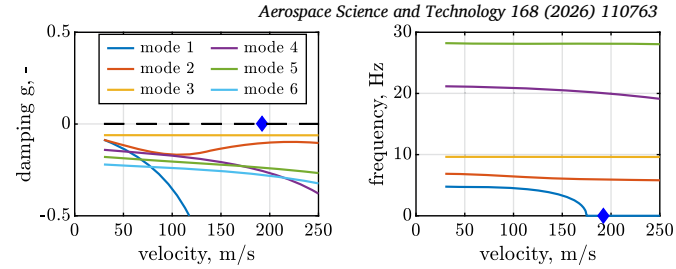
Fig. 14. Flutter speed for different combinations of η_O and η_P.

must be remembered that the minimum spacing between propellers is constrained by the requirement outlined in Section 3. A particular layout is the equally spaced DEP system, defined by placing the outermost propeller at the wingtip (i.e., $\eta_O = 1$) and setting $\eta_I = \eta_P$ (see Fig. 13b).

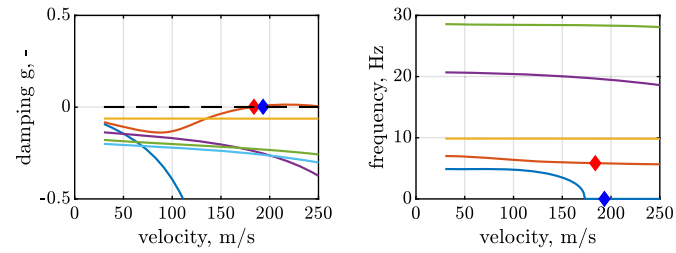
Fig. 14 presents the flutter speed computed by SDBox for the three-propeller configuration as a function of η_O and η_P. The color scale visually represents the flutter speed, with dark red indicating the lowest values and yellow denoting the highest.

The flutter mechanism consistently involves a bending-torsional coupling of the wing, leading to mode 2 crossing the zero-damping line. This behavior is expected in the nominal design conditions since wing flutter remains the dominant aeroelastic phenomenon.

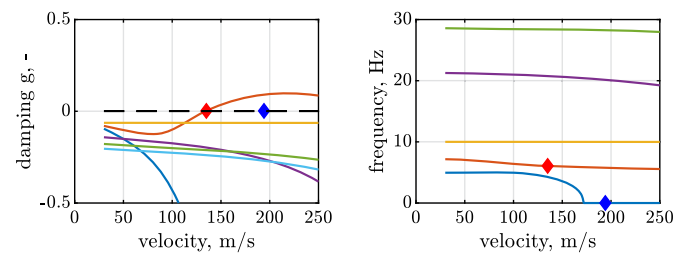
The lower boundary in Fig. 14 corresponds to the minimum allowable distance between the propellers, while the upper boundary represents the smallest spanwise position of the inner propeller. In the bottom right corner, flutter transitions to wing divergence, which occurs at $194\,\mathrm{m\,s^{-1}}$. The highest flutter velocities are observed when η_P is minimized and η_O is maximized. This indicates that the most stable configuration is achieved when the propellers are positioned close together and as near to the wingtip as possible. In contrast, the most critical configurations occur when all propellers are positioned near the wing root. For a given η_O, an increase of η_P leads to a reduction in flutter speed. Overall, the added mass from the propellers and motors lowers the natural frequencies of the coupled wing-DEP system. However, the primary factor contributing to the increase in flutter speed is the forward shift of the section's center of gravity relative to the elastic axis.



(a) Smallest distance: $\eta_P = 0.26$ & $\eta_O = 1$.



(b) Equally distributed: $\eta_P = 0.33$ & $\eta_O = 1$.



(c) Largest distance: $\eta_P = 0.43$ & $\eta_O = 1$.

Fig. 15. Flutter curves for three-propellers configuration with $\eta_0 = 1$.

The flutter diagrams for the three-propeller configuration with $\eta_0 = 1$ are presented in Fig. 15. The plots illustrate the frequency and damping ratio as functions of freestream velocity for three cases representing the smallest possible spacing between propellers ($\eta_P = 0.26$), the equally distributed spacing ($\eta_P = 0.33$), and the largest spacing ($\eta_P = 0.43$). When flutter occurs, it is driven by mode 2, coupled with mode 1. Among the three configurations, the one with the smallest propeller spacing is the most stable, exhibiting only static divergence associated with mode 1. The corresponding divergence speed is indicated in the flutter diagrams by a blue marker. At this point, mode 1 becomes real, causing one of the two roots to turn positive. Increasing the spacing between the propellers reduces stability margins and eventually triggers flutter, as indicated by the red markers in Fig. 15b and Fig. 15c. Under nominal design conditions, where the propeller dynamics remain well separated from the low-frequency wing modes responsible for flutter, no other instability mechanisms are observed.

Although these findings are demonstrated for the three-propeller configuration, the same trends are observed across other configurations.

4.2. Comparative flutter analysis of DEP configurations

This section compares the DEP configurations based on different group span positions. To limit the number of possible layouts, the distance η_P is constrained and selected to maintain an equally distributed arrangement when $\eta_O = 1$. Fig. 16 illustrates the critical speeds as a function of η_O for the five configurations. The flutter speeds denoted by the circled points correspond to the specific layout depicted on top

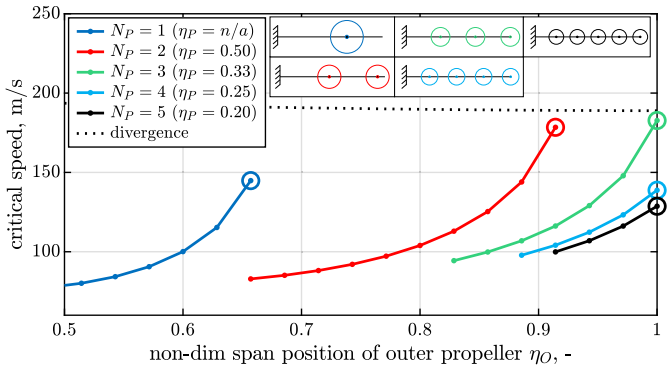


Fig. 16. Comparison of critical speeds for different DEP configurations.

of Fig. 16. Once again, the stabilizing effect of outer span positions is evident across all configurations. Configurations with more propellers exhibit lower stability margins than those with fewer propellers for the same outer propeller span position (η_O). Consequently, the single-propeller configuration is the most stable, while the five-propeller configuration is the least stable. This is primarily due to the position of the propeller and motor masses along the wing semi-span. In the single-propeller configuration, mass is concentrated at a specific span position, whereas multiple-propeller configurations distribute their masses along the span. Although additional mass at outer span positions stabilizes the aeroelastic wing-propeller system, this effect diminishes with multiple propellers, as the outer propeller-pylon becomes lighter (see Fig. 11). Notably, in the single- and two-propeller configurations, flutter is no longer detected beyond 66 % and 91 % span positions, respectively. Instead, wing divergence occurs.

Across all investigated DEP configurations, the flutter mechanism consistently involves a coupled wing bending-torsional motion, with mode 2 becoming unstable. Whirl flutter never occurs under nominal design conditions.

4.3. Aerodynamic interference on flutter for DEP configurations

Aeroelastic analyses conducted with SDBox confirm that wing flutter is the primary instability mechanism in wing-DEP systems under nominal design conditions. These conditions assume that the propeller-pylon dynamics are decoupled from the low-frequency elastic modes of the wing, which are primarily responsible for flutter and divergence. The presented outcomes are obtained by neglecting the aerodynamic interference between the propellers and the wing, with the propellers trimmed in windmilling conditions. The inclusion of aerodynamic interactions is expected to reduce the flutter boundaries. Previous investigations on a wing with a single propeller demonstrated a reduction in flutter speed due to aerodynamic interactions [35]. Furthermore, thrusting conditions exhibited lower flutter boundaries compared to windmilling, driven by the increased dynamic pressure and local lift on the wing induced by the propeller. Building on the findings of Ref. [35], the present study extends the analysis to DEP configurations, investigating the influence of aerodynamic interactions on flutter behavior using the coupled MBDyn-DUST toolchain.

The general approach follows the methodology described in [35]. The strong agreement between the flutter speeds predicted by SDBox and the MBDyn-DUST toolchain validates the use of SDBox as a reference for cases without aerodynamic interaction. The workflow employed is illustrated in Fig. 17. The SDBox model is utilized as an initial step to identify critical velocities, providing a preliminary assessment of the system's dynamic behavior. The resulting data serves as a reference for defining discrete free stream velocity intervals near the flutter speed, which are subsequently used as initial conditions in MBDyn-DUST simulations. These simulations are performed in the time domain, and the resulting time histories are post-processed using the Matrix Pencil Estimation (MPE) method to extract natural frequencies and damping ratios [42].

Table 7
Predicted flutter speeds and frequencies by SDBox, neglecting aerodynamic interaction.

	Windmilling	Thrusting
$N_P = 4$	138.8 m s^{-1}, 6.0 Hz	137.6 m s^{-1}, 6.0 Hz
$N_P = 5$	128.7 m s^{-1}, 6.1 Hz	127.8 m s^{-1}, 6.1 Hz

The trim conditions are primarily governed by the propeller's operational point. The wing's angle of attack and corresponding lift are not considered trim parameters, as it is assumed that subsonic wing flutter is predominantly driven by lift perturbations rather than the steady-state lift generated in equilibrium. The propeller operational points in both windmilling and thrusting conditions are analyzed in the following. These conditions are achieved by adjusting the collective pitch.

The study aims to offer both qualitative and quantitative insights into the influence of aerodynamic interactions on flutter onset. The critical velocities predicted by SDBox, which neglects aerodynamic interference, are compared with those extracted from time-marching simulations using the coupled MBDyn-DUST toolchain, which accounts for aerodynamic interactions. Additionally, the impact of propeller trim conditions on flutter onset is investigated.

The wing-DEP systems featuring 4 and 5 propellers are considered in the following, adopting an equally distributed layout, with the outermost propeller positioned at the wingtip. The flutter speeds and corresponding frequencies predicted by SDBox (i.e., without aerodynamic interaction) for the various configurations are summarized in Table 7. Slightly lower flutter speeds are observed in thrusting conditions compared to windmilling conditions. When aerodynamic interference is neglected, the influence of the propeller's operational point on flutter onset remains minimal, with a variation of less than 1.0%. For all configurations, the flutter mechanism is consistently driven by the coupling between wing bending and torsion.

Based on the flutter predictions, the freestream velocities for the time-marching simulations are defined. For each configuration and trim condition, at least two simulations are conducted to capture both stable and unstable dynamic responses. To achieve the desired trim conditions, the collective pitch of the propeller blades is adjusted for each freestream velocity, as shown in Fig. 18. The blue curves represent the windmilling condition, while the red curves correspond to the thrusting condition. As expected, the pitch angle required to maintain trim increases with free-stream velocity. The collective pitch angles in the thrusting case are consistently higher than those in the windmilling condition. Notably, the trim angles obtained from the coupled MBDyn-DUST simulations (solid lines) are consistently lower than those computed without accounting for aerodynamic interference (dashed lines). This offset in collective pitch is attributed to the presence of the wing, as mutual aerodynamic interference also affects propeller performance.

The system is then excited by applying a blended impulse force at the wingtip, ahead of the elastic axis, with a duration of 2 ms and an amplitude of 12 kN, ensuring sufficient excitation up to 30 Hz. The resulting dynamic aeroelastic response is recorded and post-processed using the MPE method to extract modal characteristics. An overview of the time-domain responses is presented in Fig. 19. The wingtip node displacements in heave and pitch rotations are visualized, with blue lines denoting the windmilling condition (WM) and red lines representing the thrusting condition (TH). Visual inspection highlights unstable responses at the highest freestream velocities across all configurations, while partially stable dynamic responses are observed at lower velocities. Numerical results extracted from the time histories are summarized in Table 8, where positive damping values indicate a stable response. Due to the absence of a stable response in the DEP configurations under thrusting conditions, a third simulation was necessary, reducing the free stream velocity. Linear interpolation between damping values and

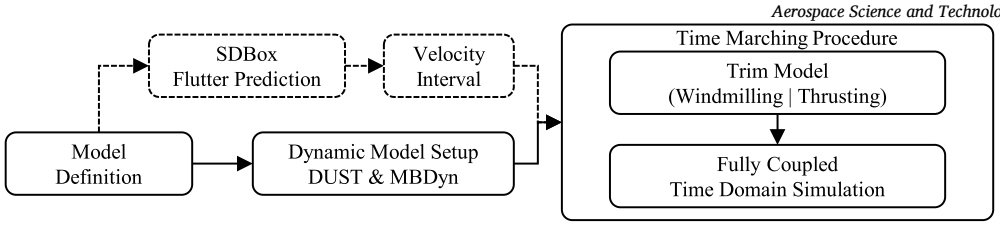


Fig. 17. Analysis workflow using SDBox and MBDyn-DUST.

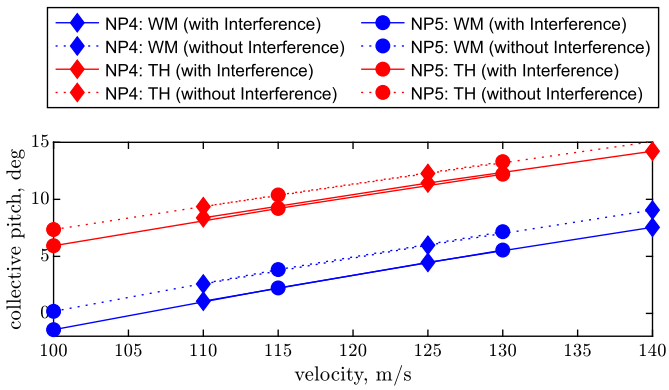
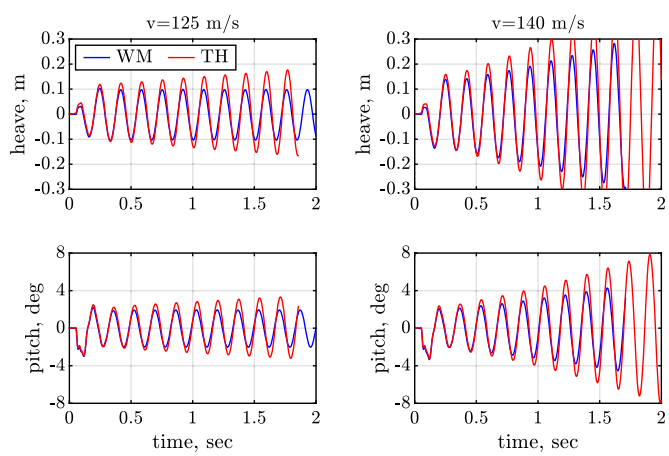
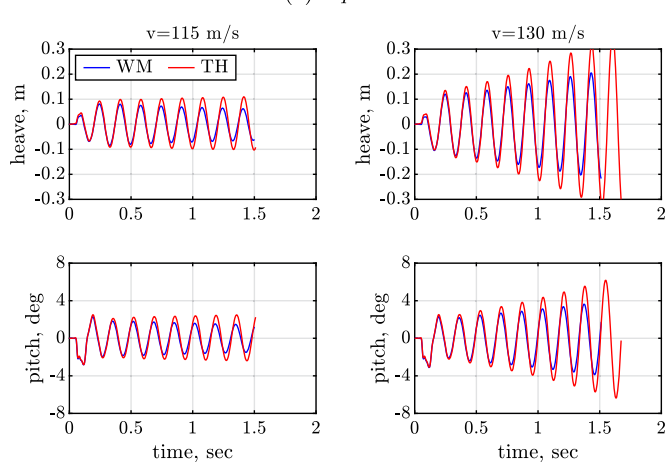


Fig. 18. Collective pitch for windmilling (WM) and thrusting (TH) conditions.



(a) $N_P = 4$.



(b) $N_P = 5$.

Fig. 19. Time histories of the wingtip node for both configurations in windmilling and thrusting conditions (MBDyn-DUST).

Table 8
Identified frequencies and damping ratios of the flutter mode for the wing-DEP configurations investigated (MBDyn-DUST).

	Freestream	Windmilling		Thrusting	
	velocity, m s^{-1}	freq. , Hz	damp., %	freq. , Hz	damp., %
$N_P = 4$	110	n/a	n/a	6.02	+1.29
	125	5.94	+0.00	5.93	-0.66
	140	5.87	-1.38	5.78	-1.57
$N_P = 5$	100	n/a	n/a	6.11	+1.52
	115	6.02	+0.53	5.99	-0.27
	130	5.91	-1.16	5.90	-0.89

Table 9
Flutter speeds (in m s^{-1}) computed by SDBox (w/out aerodynamic interference) and MBDyn-DUST (with aerodynamic interference), in windmilling and thrusting conditions.

	Windmilling			Thrusting		
	SDBox	MBDyn-DUST	Diff.	SDBox	MBDyn-DUST	Diff.
$N_P = 4$	138.8	125.0	-10%	137.6	119.9	-13%
$N_P = 5$	128.7	119.7	-7%	127.8	112.7	-12%

freestream velocities is applied to estimate the flutter speeds, as summarized in Table 9. Overall, the critical speeds predicted by MBDyn-DUST are consistently lower than those obtained from SDBox under both windmilling and thrusting conditions. The aerodynamic interaction between the propeller and the wing exhibited a marginal destabilizing effect on the flutter speed in the single-propeller configuration, with a reduction of approximately 5% (Ref. [35]). This destabilizing effect becomes more pronounced as the number of propellers increases, as the propellers' slipstream induces a higher dynamic pressure over a larger wing area in DEP configurations (see Fig. 20). The effect is particularly significant under thrusting conditions, where the propellers generate higher induced velocities compared to the windmilling scenario.

The coupled MBDyn-DUST simulations also highlight the stronger influence of the propeller trim condition on flutter predictions. The results (see Table 9) confirm that thrusting conditions yield more conservative flutter speeds compared to windmilling, with aerodynamic interactions playing a crucial role in the destabilizing effect.

While frequency-domain methods remain more efficient for rapid flutter predictions, the present findings suggest that such methods tend to underestimate flutter boundaries in wing-propeller systems — particularly in DEP configurations — due to the absence of aerodynamic interaction effects. Therefore, higher-fidelity methods are essential to improve accuracy in these cases. Nonetheless, frequency-domain methods are still recommended as a first step to quickly identify the velocity range where flutter is likely to occur and to characterize the nature of the flutter mechanism. This information can be leveraged to minimize the number of high-fidelity simulations required to refine the flutter boundaries.

5. Design guidelines

Based on the literature review and the findings from the authors' recent research, aircraft with wings equipped with a DEP system may

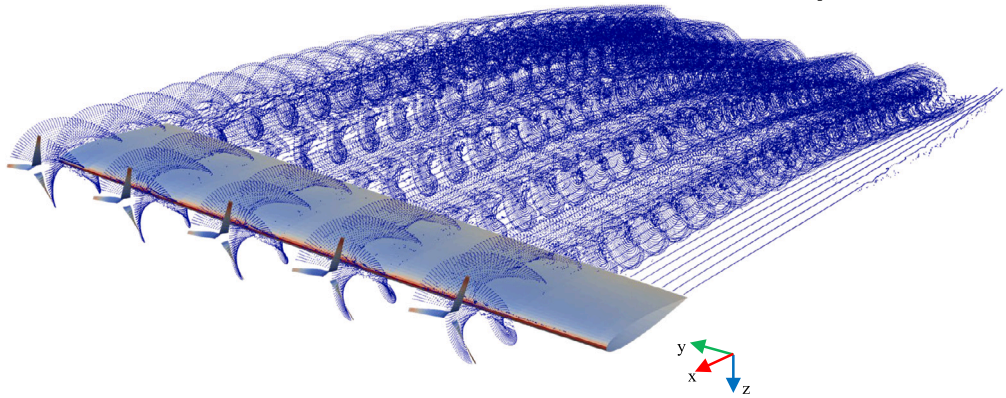


Fig. 20. DUST simulation snapshot of the five-propeller configuration with particle visualization ($V_\infty = 130$ m/s).

encounter the following aeroelastic instabilities, ranked by likelihood of occurrence:

1. Wing flutter;
2. Wing divergence;
3. Whirl divergence;
4. Whirl flutter.

Under normal operating conditions, wing flutter is the primary aeroelastic instability. The dynamics and aerodynamics of the propeller-pylon systems may play a significant role in influencing this behavior. Indeed, a well-designed wing-DEP system can substantially stabilize or even eliminate wing flutter. In such cases, wing divergence then becomes the dominant instability mechanism.

A low mounting stiffness between the pylon and wing theoretically promotes whirl flutter. However, this scenario is unlikely in practice, as it would require similar stiffness and modal mass in both pitch and yaw directions.

If the pylon is rigidly attached to the wing, whirl flutter would need to be triggered solely by the wing's elastic behavior. Pitch and yaw motions would be driven by wing torsion and in-plane bending, respectively. In such cases, low torsional stiffness combined with a higher torsional modal mass (contributed by the wing) would likely make wing divergence the critical instability.

Whirl flutter is also unlikely in configurations with tilting mechanisms controlled by actuators. These configurations typically have different stiffness distributions in pitch and yaw. Pitch stiffness, mainly determined by the actuator, is generally lower than yaw stiffness due to the wing's in-plane compliance. The corresponding modal masses also differ, further reducing the likelihood of whirl flutter. However, in cases where the actuator's pitch stiffness is weak, propeller-pylon divergence may become a potential instability.

The following presents design guidelines for mitigating aeroelastic instabilities in aircraft equipped with DEP systems. These guidelines aim to enhance safety and reliability by reducing or potentially eliminating aeroelastic phenomena. Implementing these measures may impact other design factors, such as increased system weight or reduced overall performance. Therefore, the proposed strategies should be applied judiciously and only when aeroelastic instabilities are anticipated within the aircraft's flight envelope. Ultimately, aircraft design is a continuous trade-off between competing disciplines, requiring careful balance to achieve optimal performance and safety.

The design guidelines are divided into two subsections: the first focuses on mitigating aeroelastic instabilities related to the propellers, while the second addresses instabilities in wings equipped with a DEP system.

5.1. Guidelines for propeller-pylon systems

The flexibility of the pylon plays a significant role in the occurrence of whirl flutter in propeller-pylon systems. The most critical scenarios arise when the stiffness and modal masses in pitch and yaw are equal or nearly identical [17].

As a general guideline, increasing structural stiffness can help mitigate whirl flutter risks. However, failure scenarios should be addressed during the certification process, particularly when the likelihood of structural failure and the resulting loss of stiffness exceeds a critical threshold. As a result, design parameters are constrained by the minimum available stiffness under failure conditions and the operational range of the aircraft.

A summary of the key parameters affecting propeller-pylon stability is presented below.

5.1.1. Distance between pivoting point and propeller hub

As demonstrated by Houbolt et al. [16], increasing the distance between the hub and the pivot point enhances whirl flutter stability margins. This stabilizing effect is primarily attributed to the aerodynamic benefits of the longer lever arm. However, increasing this distance also raises the second moment of inertia of the propeller-pylon system about the pivot point, particularly when heavy masses are concentrated near the propeller hub. In cases where an electric motor is used—especially if it is directly mounted at the hub—the mass properties may negatively impact aeroelastic stability, potentially offsetting the aerodynamic stabilizing effect.

Fig. 21 illustrates the required pylon stiffness for whirl flutter stability under isotropic conditions (i.e., $K_\theta = K_\psi$) as a function of the hub-to-pivot distance and various motor masses, using the propeller #1 considered in this study. The destabilizing effect of increased inertia at the pivot point due to a heavier motor is clearly evident. Additionally, a peak in the required stiffness rises as motor mass increases. Beyond this critical point, the system can be stabilized by extending the pylon length.

From a design standpoint, the motor must be selected to provide the required power. When multiple motor options are available, selecting the lightest one is ideal for improving whirl flutter stability. However, this recommendation is specific to increasing whirl flutter boundaries. As discussed later, a heavier motor mounted near the wing's leading edge can enhance wing flutter stability.

In general, maximizing the distance between the propeller hub and the pivot point improves aeroelastic stability. However, longer pylons also increase the bending moments transmitted to the pivot point. The use of shock mountings for the motor may be considered as a potential solution, but it requires careful evaluation, especially when the motor is positioned near the propeller hub. These mountings introduce additional flexibility, which can create a secondary pivot point. This, in turn,

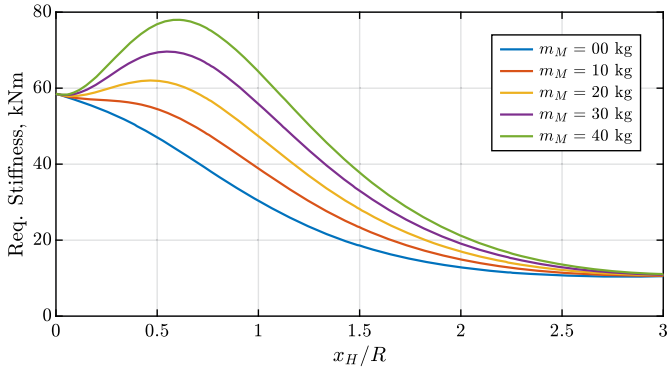


Fig. 21. Whirl flutter boundary based on the distance between pivoting point and hub for different motor masses.

reduces the hub-to-pivot distance, potentially compromising whirl flutter stability margins [43,44].

5.1.2. Propeller size

The selection of propeller size, particularly the radius, primarily considers aerodynamic and performance factors. Changes in radius directly affect thrust, torque, and efficiency, thereby altering operating conditions and influencing the whirl flutter speed. A multidisciplinary approach to propeller design is most effective when aiming to maintain consistent performance (e.g., total thrust) without compromising aeroelastic stability [38,45]. For instance, when multiple propellers are used to replace a single propeller in a given configuration, the thrust requirement for each propeller must be adjusted accordingly. Larger propellers, along with their corresponding electric motors, are generally more susceptible to whirl flutter due to increased mass and stronger gyroscopic effects [46]. This susceptibility can be mitigated by the greater elasticity of larger propeller blades, which has a stabilizing effect, as discussed in Ref. [19].

5.1.3. Propeller trim condition

The propeller trim condition significantly influences whirl flutter boundaries [24]. For a given thrust requirement, higher rotational speeds combined with a smaller collective pitch of the propeller blades were found to increase the whirl flutter speed. This combination increases the $c_{m,q}$ propeller stability derivative, representing the change of pitch moment at the propeller hub due to the pitch rate, which has a stabilizing effect on whirl flutter, as discussed in Ref. [21]. However, it should be noted that this configuration also demands more power, reducing overall propeller performance and necessitating a heavier motor.

5.1.4. Aerodynamic torque and recuperation propellers

A further stabilizing effect on whirl flutter is obtained by the aerodynamic torque when considering powered propellers [24]. Special care should be taken for propeller systems with recuperation capabilities [47,48] as the torque reverses its direction, requiring higher mounting stiffness to maintain adequate stability margins.

5.2. Guidelines for wings with single and multiple propellers

The following design guidelines provide a framework for optimizing wing-DEP systems, primarily aimed at increasing wing flutter boundaries, the most critical mechanism under nominal design conditions. Where relevant, the guidelines will also address the influence on whirl flutter stability. These recommendations are applicable to any number of propeller-pylon systems mounted on the wing, making them suitable for both single-propeller and multiple-propeller (DEP) configurations. Additionally, a comparison and recommendation regarding the optimal number of propeller-pylon systems will be presented at the end of this section.

5.2.1. Propeller-pylon attachment along the wing chord

The position of the propeller-pylon system along the wing chord significantly influences the center of gravity (CoG) axis of the wing-DEP system, making it a critical factor for aeroelastic stability. A forward-shifted CoG axis relative to the elastic axis is known to enhance wing flutter stability by reducing inertial and aerodynamic coupling [10]. However, placing a single lumped mass ahead of the elastic axis primarily shifts the CoG locally (i.e., only at the semi-span location of the propeller-pylon system) and might result in only minor improvements in flutter stability compared to a clean wing [14]. The effect is most pronounced when the lumped mass is placed near the wingtip, as discussed in the next section.

From an aeroelastic perspective, the tractor configuration is the most stable for wing-DEP systems and is generally preferred over the pusher configuration. A hybrid configuration with both tractor and pusher propellers may also be viable, but it requires careful assessment of its impact on the overall CoG axis. The potential stabilizing effect of the tractor propellers' added mass could be offset by the destabilizing effect of pusher propellers, especially since the distances between the propeller hubs and the wing's leading edge are typically shorter than those between the trailing edge and rear-mounted hubs.

The most significant drawback remains the increased overall mass of the aircraft, resulting in performance reduction. A longer lever arm between the elastic axis and the propeller-pylon system (i.e., using a longer pylon) may offer a more practical stabilization method. This configuration also helps improve whirl flutter stability, though it also results in higher moments being transmitted from the propeller to the wing.

5.2.2. Propeller-pylon attachment along the wing span

The spanwise positioning of propeller-pylon systems plays a crucial role in stabilizing the aeroelastic behavior of the wing, with stability increasing as the propellers are positioned closer to the wingtip, where modal displacements and rotations are higher. For single-propeller configurations, a wingtip-mounted propeller is the most effective arrangement for mitigating wing flutter [14]. In distributed electric propulsion (DEP) configurations, the present study shows that the most stable layout is achieved by clustering the propellers as closely as possible near the wingtip. This setup also offers aerodynamic performance benefits [49]. However, it can lead to increased bending moments at the wing root, a factor that must be carefully addressed during the design process.

5.2.3. Coupling propeller-pylon and wing dynamics

This scenario arises when the natural frequencies of the propeller-pylon system fall within the low-frequency range of the wing's elastic modes responsible for flutter. Several design parameters influence the natural frequencies of the propeller-pylon system, including propeller mass, inertia, rotational speed, distance between the hub and pivot point, and the pylon's equivalent stiffness. Among these factors, low natural frequencies are most likely to occur in failure conditions where the pylon stiffness is reduced.

A substantial reduction in pylon mounting stiffness can induce propeller whirl flutter. In such cases, the backward frequency may drop below the wing's first elastic frequency, initiating a localized instability that can propagate throughout the entire system. However, due to the stabilizing influence of the elastic wing, whirl flutter speeds are generally expected to exceed the critical speeds predicted for the standalone propeller-pylon system [17].

Failure conditions in which the propeller's natural frequencies fall within the low-frequency range of the wing's elastic modes may lead to coalescence phenomena and resonance effects, potentially reducing the wing flutter speed or triggering additional instability mechanisms. Unfortunately, a generalization of these phenomena is not possible, as the aeroelastic behavior of wing-DEP systems is influenced by numerous factors. To identify the worst-case scenario, parametric analyses must be performed for each specific configuration by varying the pylon mounting stiffness of each propeller. In this context, frequency-domain

methods are the most suitable approach to efficiently cover the large test matrix required for flutter clearance. As a reference example, the results of the failure analyses for the five configuration layouts investigated in this work can be found in [50].

Finally, it is worth noting that, in certain cases, propeller-pylon dynamics can have a stabilizing effect by increasing the flutter speed. This occurs when the propeller-pylon system couples with one of the two wing modes driving flutter, resulting in an increased frequency gap between the two modes. As a consequence, the onset of flutter is delayed, leading to a higher critical flutter speed [50].

5.2.4. Selecting the number of propellers in DEP systems

This study demonstrates that increasing the number of thruster propellers, while maintaining the same thrust requirement, disk loading, and total mass lowers the flutter boundaries. This reduction is primarily attributed to the position of propeller and motor masses along the wing span. As the number of propellers increases, smaller masses are positioned near the wingtip, reducing the stabilizing effect of having the center of gravity ahead of the elastic axis in the wing's outer regions, where modal displacements and rotations are more pronounced. Additionally, the increased number of propellers intensifies the aerodynamic interaction between the propellers and the wing, generating higher dynamic pressure regions, which has been shown to lower stability margins. However, DEP systems also provide aerodynamic benefits, particularly by enhancing wing performance [6,7,51]. That said, the number of propellers cannot be increased indefinitely. Maintaining a uniform disk load imposes design constraints on the propeller blades. As the number of propellers increases, the thrust requirement per propeller — and consequently its diameter — decreases. This, in turn, necessitates smaller and thinner blade chords, which could result in impractically small propeller blades (see Section 3).

In conclusion, achieving the optimal design of a wing-DEP system requires a careful trade-off between various factors. This trade-off should consider aeroelastic stability and performance as well as practical considerations such as the availability of motors with suitable mass and the increased maintenance costs associated with a larger number of components. Balancing these factors is essential to determine the optimal number of propellers for the specific configuration. An aeroelastic optimization method for the preliminary design of DEP wing laminates, accounting for propeller whirl flutter effects, was recently developed by Wang et al. [52]. However, additional tools tailored for eVTOL aircraft design are anticipated, offering enhanced capabilities to integrate aeroelastic constraints.

Finally, the integration of the flight control system should be considered as a critical element within a multidisciplinary design framework. Active control strategies offer a promising means to suppress flutter and whirl flutter instabilities by leveraging the distributed control authority inherent in DEP configurations. This distributed thrust capability can be harnessed to provide active damping and mode suppression, thereby enhancing overall aeroelastic stability [53,54].

6. Conclusions

This work analyzes the aeroelastic instability mechanisms across different wing-DEP configurations. The wing is representative of a mid-range small aircraft, similar to an eVTOL. Configurations with up to five thruster propellers per wing semi-span were analyzed. Nominal design conditions were considered, maintaining an adequate frequency separation between the natural frequencies of the isolated propeller-pylon system and the wing's elastic modes critical for flutter. The number and positioning of the propellers along the wing span were varied to identify potentially advantageous configurations for flutter clearance.

The primary instability mechanism identified was flutter, driven by the coupling of wing bending and torsion. The analysis revealed that the flutter mechanism remained consistent across all configurations, regardless of propeller placement. The most stable configuration was achieved with a single propeller positioned at the wingtip. Overall, as the number of propellers increased, the flutter speed decreased. However, more stable layouts were observed when the propellers were placed close together near the wingtip. This stabilization effect stems from the propeller and motor masses shifting the center of gravity of the wing-DEP system ahead of the elastic axis. The effect is more pronounced when these masses are positioned near the wingtip, where modal displacements and rotations are highest. It is important to note that while shifting mass at the wingtip increases the aircraft's aeroelastic stability, it also leads to an increase in the predicted root bending moment. Therefore, selecting the number of propellers and their placement along the wing span requires a careful trade-off analysis to identify the optimal configuration that minimizes wing root loads while maximizing flutter boundaries.

Aerodynamic interactions contribute to a further reduction in flutter speed, with a significant impact observed in thrust conditions. In relative comparison to windmilling conditions, configurations with multiple propellers exhibit a higher degree of destabilization. This trend is likely the result of higher local dynamic pressures caused by the propeller slipstream. The larger affected area with multiple propellers likely amplifies this destabilizing effect.

Finally, design guidelines are provided to enhance the stability of both propeller-pylon and wing-DEP systems. These recommendations aim to improve safety and reliability by mitigating or eliminating aeroelastic phenomena. However, implementing such measures may introduce trade-offs, including increased system weight or reduced overall performance. A multidisciplinary approach is therefore essential to identify the best configuration for next-generation sustainable aircraft.

The design and verification of wing–DEP systems will ultimately require dedicated experimental campaigns to corroborate the findings of numerical aeroelastic analyses. Although several experimental studies have investigated wing–propeller systems to assess performance variations due to aerodynamic interference, these efforts have largely relied on rigid structural models. To capture the coupled aeroelastic behavior inherent to wing–DEP configurations, such rigid components must be replaced with elastic elements. Developing a representative wind tunnel model will necessitate a carefully defined scaling strategy to concurrently preserve both the structural dynamics and aerodynamic characteristics of the full-scale system. In addition, the test setup should allow for configurational adaptability, such as varying the number and spanwise distribution of propellers, to enable systematic parametric investigations.

CRediT authorship contribution statement

Nils Böhnisch: Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Carsten Braun:** Writing – review & editing, Validation, Supervision, Data curation, Conceptualization. **Pier Marzocca:** Writing – review & editing, Validation, Supervision, Data curation, Conceptualization. **Vincenzo Muscarello:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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